

Aptitudinal Alignment Quantum Manual

Institutional Fidelity Field Theory

Aptitude Systems

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Listings

1 Introduction

Aptitudinal Alignment operates at the quantum level of institutional behavior. Each public record, decision, or procedural act is treated as a discrete quantum of agency behavior. These quanta interact through measurable fields of fidelity, consistency, transparency, efficiency, equity, and accountability. The emergent field defines the Aptitude Score — a scalar representation of institutional alignment with the public trust.

2 Quantum Ontology of Agency Behavior

2.1 Fundamental Entities

- **Behavioral Quantum** (q_b): smallest measurable unit of agency action.
- **Signal Particle** (s_i): observable manifestation of a behavioral quantum (e.g., record entry, decision, correction).
- **Field Tensor** (Φ_{ij}): interaction matrix between behavioral quanta across dimensions.
- **Alignment Wave** (Ψ_A): emergent probability field representing institutional fidelity.

2.2 Quantum Interaction Equation

Each agency's behavior field evolves according to:

$$\frac{\partial \Psi_A}{\partial t} = \nabla \cdot (\Phi_F + \Phi_C + \Phi_T + \Phi_E + \Phi_Q + \Phi_A)$$

Where each Φ_X represents the field tensor for one Aptitude dimension.

3 Data Ontology and Event Structure

3.1 Agency Event Object

```
{
  "agency_id": "uuid",
  "event_id": "uuid",
  "event_type": "procedural_action",
  "quantum_signature": {
    "fidelity": 0.91,
    "consistency": 0.87,
    "transparency": 0.76,
    "efficiency": 0.82,
    "equity": 0.68,
    "accountability": 0.74
  },
  "timestamp": "2026-06-24T18:44:00",
  "source": "public_record"
}
```

3.2 Quantum Relational Schema

```
CREATE TABLE behavioral_quanta (  
  id UUID PRIMARY KEY,  
  agency_id UUID,  
  event_id UUID,  
  dimension TEXT,  
  magnitude FLOAT,  
  timestamp TIMESTAMP  
);
```

```
CREATE TABLE field_interactions (  
  id UUID PRIMARY KEY,  
  from_quantum UUID,  
  to_quantum UUID,  
  field_type TEXT,  
  interaction_strength FLOAT  
);
```

4 Quantum Scoring Model

The Aptitude Score emerges as the expectation value of the alignment wave:

$$A = \langle \Psi_A | \hat{F} + \hat{C} + \hat{T} + \hat{E} + \hat{Q} + \hat{A} | \Psi_A \rangle$$

Where each operator $\hat{X}$ acts on the behavioral quanta corresponding to its dimension.

5 Dimensional Field Definitions

- **Fidelity Field (Φ_F):** measures adherence to mission and legal mandate.
- **Consistency Field (Φ_C):** measures variance in outcomes across similar cases.
- **Transparency Field (Φ_T):** measures visibility and accessibility of records.
- **Efficiency Field (Φ_E):** measures conversion of resources into outcomes.
- **Equity Field (Φ_Q):** measures fairness across demographics and geographies.
- **Accountability Field (Φ_A):** measures corrective and self-audit behavior.

6 Agent Architecture Integration

Each agent in the Aptitude Resolution Layer operates as a quantum processor:

- **Intake Parser Agent:** converts raw public data into behavioral quanta.

- **Legal Classifier Agent:** assigns quanta to doctrinal categories.
- **Signal Harmonizer Agent:** normalizes magnitudes across jurisdictions.
- **Field Synthesizer Agent:** computes interaction tensors.
- **Alignment Evaluator Agent:** collapses the wave function into a scalar Aptitude Score.

7 Governance and Ethics Field Constraints

The Aptitude field obeys conservation laws:

$$\frac{d}{dt}(\Phi_F + \Phi_C + \Phi_T + \Phi_E + \Phi_Q + \Phi_A) = 0$$

Meaning institutional fidelity cannot be created or destroyed — only transferred through transparency and accountability.

8 Design Principle

Aptitude is a civic physics engine. It does not judge individuals; it measures the coherence of institutions with the public trust. Every agency emits behavioral quanta; Aptitude measures their interference pattern to reveal the true alignment field.